- Companies (Amendment) Act, 2002 is expected to enhance the competitiveness of cooperatives, and enable them to compete and operate in today's liberalised, globalised market.

- In Companies (Second Amendment) Act, 2002 industrial sickness has been redefined; a revival and rehabilitation fund has been set up; protection from creditors has been withdrawn.

- The Competition Act, 2002 aims at promoting competition through prohibition of anti-competitive practices, abuse of dominance and through regulation of companies beyond a particular size.

- DIN (Director Identification Number) has been issued to all the directors of all the companies under a surveillance system. The system will give the government instant access to the details and nature of employment relevant to company law requirements and antecedents which are crucial to investor protection.

The internal liberalisation has been accompanied by a policy of maintaining an *open access to imports* to permit modernisation and technological upgradation in Indian industry which is expected to reduce costs and promote international competition.

The *aim of sweeping policy changes* is to evolve an integrated economic package that can be implemented in stages to create an appropriate environment so as to encourage and promote greater efficiency, higher productivity and faster industrial growth in desired directions through a well-coordinated system of incentives. Accelerated growth of manufacturing, accompanied by radical restructuring and induction of 'sunrise' industries within a suitably modified policy frame would bring about a significant transformation of India's industrial economy.

Positive Effects

India has woken up to the liberating influence of what Joseph Schumpeter called "creative destruction"– the death of the outdated at the hands of the modern. The following can be identified as some of the positive effects:

1) The post-liberalisation era has propelled companies into a restructuring spree; companies are doing away with their not-so-profitable businesses. Corporate sell offs are moving at a pace never seen before.

2) More ambitious players have been consolidating themselves by way of mergers and acquisitions. Entrepreneurs have come to believe: "In business, big is always beautiful."

3) Over the last two decades, Indian manufacturing companies have emerged on a par with the best in the world from the quality perspective. This has happened because Indian manufacturing has adopted world-class practices in manufacturing management by educating their employees, both managers and shop-floor staff with the help of global teachers, mainly Japanese, who have brought in the best manufacturing management techniques.

4) India's share in world market capitalisation is now more aligned with its share in global GDP.

5) Economic reforms have created an environment conducive for low-cost innovation. This, combined with increasing pressures for inclusiveness, will contribute to creating an income-pyramid that will develop a bulge in the middle– faster than earlier forecast. That is creating a huge mass market for goods and services at a price that is affordable for this segment.

6) Indian business has emerged leaner, more efficient in terms of process, quality and financing, and becoming competitive on a global scale. Indian companies today are expanding operations in overseas markets through both organic and inorganic means. There is a sense of optimism, and the ability to think big and execute large plans. In addition, companies have developed the ability to quickly respond to changes in market conditions. For example, in response to the recent global economic slowdown, they aggressively reduced their inventories, realigned production levels, and cut costs to rebase to the new cost price-demand equation.

7) Companies are increasingly going in for *coopetition*, *i.e.*, getting together to become more competitive.

8) Trade unions and workers have responded positively to the economic reforms. Their open-minded approach towards adoption of new technologies and productivity linked wage agreements would go a long way in consolidating the future of Indian industry.

9) Liberalisation has altered the investment behaviour of Indian entrepreneurs. Industrialists have fast realised the role of scale economies, rapid technological growth and increased productivity. Indian companies are now going in for world-size plants. This will enable them to meet the competitive challenge of MNCs. Many Indian companies, for the first time, have crossed the billion-dollar mark in annual turnover.

10) The restructuring process of the corporate sector has gained momentum with foreign collaborators seeking to enhance equity in the Indian ventures, to gain a foothold in the management. The money comes in with strings attached: board membership, due diligence and even some operational oversight.

In short, liberalisation has opened up a new era which stresses the importance of both economies of scale and quality of products; these hold the embryo of higher productivity and competitiveness both in the home market and the export markets, only if the Indian industry responds positively to the challenge thrown to it.

Negative Effects

From a force that unleashed India's creative energies, markets are increasingly seen as an institution that seems emblematic of homo homini lupus – man is wolf to another man - and capitalism's genius of "creative destruction" appears in popular discourse as a force that is more destructive and less creative.

The current phase of investment following liberalisation also need to be given a second look. While substantial investments might have been flowing into a few industries the Government has reportedly expressed its concern over slow pace of investments in a few basic and strategic industries such as engineering, power, machine tools, etc. It has often been pointed out that the rate of return in these sectors is not more than the expectations in the newer or 'Sunrise' areas. Thus, the rate of growth of the industries wherein sufficient investment is not coming could slacken. Such distortions in the investment pattern need to be rectified for the sake of balanced growth of industrial economy.

Need for Strengthening Interlinkages between New and Old Sectors

New sectors should have strong linkages with the old ones and should push up the latter towards modernisation and new product development. Unless interlinkages are strengthened, a part of impetus given by the new sectors could be lost through leakages abroad.

New Game-Changers

Indian investors tracking basic industries had a relatively easier task till now. They had to keep one eye on the macroeconomy and assess the interplay of factors such as demand-supply, raw material cost and availability, and the landed cost of imports. These factors can be tracked since they are tangible and measurable.

Government Policy has now emerged as a new and significant factor on the horizon, one which is intangible and, hence, not easily measured. Environmental policy and regulations at the central and state levels are affecting business. The key players are the government itself, regulators (such as pollution control boards) and courts.

14.4.2 Recent Policy Initiatives Impacting Industrial Growth

Implementation of GST

The GST is a game changing reform introduced by the government. It is expected that implementation of GST will facilitate the creation of one common market in the country by removing tax barriers; eliminate cascading of taxes thereby reducing cost of production of manufacturing goods; and enhance ease of doing business by cutting down transaction costs associated with the complex tax regime. The implementation of GST is also going to cover the unorganised sector industries.

Make in India

The 'Make in India' programme has been launched globally on 25th September 2014 which aims at making India a global hub for manufacturing, research and innovation and integral part of the global supply chain. This initiative is based on four pillars of New Processes, New Infrastructure, New Sectors and New Mindset.

Start-up India

Start-up India is a flagship initiative of the Government of India, intended to build a strong eco-system for nurturing innovation and Start-ups in the country that will drive sustainable economic growth and generate large scale employment opportunities. The Government through this initiative aims to empower Start-ups to grow through innovation and design.

Ease of Doing Business

The Government has taken up a series of measures to improve Ease of Doing Business. The emphasis has been on simplification and rationalisation of the existing rules and introduction of information technology to make governance more efficient and effective. States too have been brought on board in the process to expand the coverage of these efforts.

Intellectual Property Rights (IPR) Policy

In May, 2016, Government for the first time adopted a comprehensive National Intellectual Property Rights (IPR) policy to lay future roadmap for intellectual property. This aims to improve Indian intellectual property ecosystem, hopes to create an innovation movement in the country and aspires towards "Creative India; Innovative India".

Check Your Progress 2

1) Outline the main features of strategy for the industrial sector pursued post 1991 reforms.

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2) Compare and contrast the positive and the negative effects of liberalisation on the Indian industrial sector post 1991 reforms.

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3) Outline the recent policy initiatives impacting the Indian industrial sector.

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14.5 CRITICAL ISSUES BEFORE INDUSTRIAL SECTOR

14.5.1 Industrial Sickness

India is probably unique among market economies in having policies which override the normal processes of adjustment and which effectively prevent firms from closing down their operations. The result has been a widespread and growing incidence of the special Indian phenomenon of 'Industrial sickness'.

Definition of Sickness

The definition of sickness has undergone changes over the years. The latest definition of sick units is such that the related default of the loan amount is enough to categorise it as a sick unit. Technically speaking, the loan account of the unit has to become a non-performing asset (NPA).

Causes of Sickness

The important causes of industrial sickness can be classified as: (1) external causes, and (2) internal causes.

External Causes

i) High costs of manufacture coupled with a low realisation of sales revenue. High costs may be due to inflated prices of inputs. A low sales revenue may be accounted for by lack of control over prices of output.

ii) Non-availability of raw materials, regularly and smoothly, or availability at high prices.

iii) A lack of regular supply of inputs such as power and transport bottlenecks.

iv) A general recessionary trend in the economy affecting the overall performance of industrial units. This will be reflected in a downward sloping demand curve or lack of demand altogether, particularly for industries which have a derived demand.

v) Fiscal imposts such as excise duties, import duties, etc. These more generally undermine the profit margins.

Internal Causes

i) An improper demand estimation for the products to be sold. Normally, only industry-wise demand estimations are made without ascertaining the particular factors which account for a specific demand for a particular unit's production.

ii) An improper choice of technology, unsuitability of product mix, or single product technology, wrong location of industry, non-flexibility of fixed assets, mainly machinery, for possible use in the diversified manufacturing set-up.

iii) A defective capital structure specially on account of delayed constructions and operations, resulting in cost overruns and larger borrowings. Moreover, an inability to raise adequate financial resources to with-stand operational losses and bear their impact, in the initial stages, due to a poor equity base, will appear to be a severe constraint.

iv) A growing shortage of working capital, as the units go into operation, due to a shortage of raw materials, and high prices, poor debtors' collection, inadequate inventory management, etc. are serious constraints.

v) Managerial ineffectiveness, poor control and absence of control on such key areas of operations as finance, inventory and marketing.

Of all the factors mentioned above, it is the mismanagement[1] that has been the most important cause of sickness.

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[1] Until recently, under what Raj Krishna called the *dharmshala* model (a shelter for the pious poor), government guaranteed capital against failure by taking over sick industries.

Government Policy

The Government in co-operation with the RBI have instituted arrangements for monitoring sickness of industrial units so that corrective action is taken in time. The legislative and institutional framework for dealing with industrial sickness is contained in the Sick Industrial Companies Act, (SICA) 1985 that provided for the setting up of the Board for Industrial and Financial Reconstruction (BIFR).

Rehabilitation

The rehabilitation programme involves the following major issues which have to be sorted out after it is decided that a sick unit is viable and should be rehabilitated: (a) Change of management, (b) Development of a suitable management information system, (c) A settlement with the creditors for payment of their dues in a phased manner, taking into account the expected cash generation as per viability study, (d) Determination of the sources of additional funds needed to refinance, (e) Modernisation of plant and equipment or expansion of an existing programme or even diversification of the products being manufactured, (f) Concession or reliefs or assistance to be allowed by the state level corporation, financial institutions and Central Government.

14.5.2 Technological Obsolescence and Modernisation

Another related problem Indian industry is faced with is that of technological obsolescence a large segment of industry. As a result, input intensity has increased, productivity is low, costs high and quality poor. All this means losing out in the world market and cutting into the rate of growth of income. Indian industry needs modernisation, which is a multidimensional concept. It may cover replacement of an obsolete machinery by a new machinery. A basic test of modernisation is that it should lead to a substantial reduction in the unit cost of production

Causes of Slow Pace of Modernisation

The basic cause has been the paucity of finance resulting from the following major handicaps:

1) ***Licensing Policy***. Sometimes, modernisation of a unit may bring about expansion of capacity. In the past restrictive industrial policy was followed, whereby expansion and modernisation proposals, specially from the large houses, were rejected because these led to capacity addition.

2) ***Price Controls***. Price controls tend to depress the rate of return. This prevents industry from retaining adequate profits which can be deployed to replace machines in time. From experience, it is found that even when costs of production rise sharply, the adjustment in prices follow invariably with a time lag, resulting in substantial erosion of funds for replacement and modernisation.

3) ***Depreciation Provisions***. Depreciation provisions based on historical cost of acquisition of machinery are inadequate to replace machinery in a period of rising costs and prices. Industry is forced to borrow just to keep the production capacity intact. Modernisation, consequently, is delayed.

4) ***High Rate of Corporate Taxation***. This leaves industry with too little earnings for modernisation. Although recently (2019) this has been brought down to regionally competitive levels.

The combined effect of the above factors has been the erosion of funds available for replacement or modernisation.

Suggested Measures

To put the modernisation programme through, certain policy modifications will be required. These can be classified in two categories: (a) those required to meet immediate needs, and (b) those over time.

Among the first is the need to make available more and cheaper funds through the financial institutions. Viability of an enterprise has a be the primary and the most important criterion for lending these funds. There should not be insistence on any specific debt-equity ratio or promoter's contribution. Similarly, investment allowance and excise rebate on production could be granted.

For future needs, following steps can be suggested:

i) Units may be allowed to set aside a percentage of profits and depreciation towards modernisation reserves. This presumes that units have adequate profits.

ii) In computing the cost of production for fixing prices, modernisation requirement must be taken into account.

iii) For purposes of taxation, units may be allowed to carry back losses against profits of the previous three years.

iv) The import policy for capital goods may be modified. The possibility of tying up modernisation with export production can also be considered.

v) Developments in technology are taking place across the globe. Government and industry must work together to complete the programme within the shortest possible time.

14.5.3 Productivity in Indian Industry

Productivity is a measure of performance of the production activity and refers to the amount of output produced per unit of input. It is possible to express this in the following equation:

Here, output stands for a weighted sum of various products, whereas input stands for a weighted sum of various inputs. So defined, the concept of productivity refers to the total factor productivity, rather than the partial measures of productivity like labour productivity (*i.e.*, output per unit of labour); and capital productivity (*i.e.*, output per unit of capital). The partial productivity indices are dominantly influenced by the process of capital deepening (or increasing capital-labour ratio) which is normally associated with the process of capital accumulation. It is, therefore, necessary to go beyond the partial factor productivities to analyse the total factor productivity growth (TFPG).

Importance of Productivity

For a country like India, with a multiplicity of socio-economic demands on its capital, how the limited resources are utilised assumes importance. While substantial improvements in production process can be a precondition for economic transformation, it is only the productivity of investments undertaken which yields further re-investible resources. These generate surpluses, which then motivate entrepreneurs toward undertaking further industrial activity.

Secondly, productivity growth also comes in handy in an attempt to enhance the competitiveness of a country's exports. Productivity growth lowers labour costs and thus, *ceteris paribus*, the international price of the good concerned.

Thirdly, TFP is now accepted as the main contributing factor of economic growth. The central idea is that due to the law of diminishing returns, increased use of inputs simply fails to yield increased output in the long run. Sustained output growth requires not so much the dollops of capital as technological sophistication, managerial innovativeness and shop floor acumen.

Fourthly, productivity growth can have a positive impact on poverty reduction through two channels. *One*, an increase in productivity potentially raises wages and incomes, and hence reduces poverty. Higher productivity-led wages and incomes can have a second-round impact on domestic demand, and in turn, on employment and in further gains in poverty reduction. *Two*, productivity gains help to moderate the rate of increase in prices. Lower inflation is equivalent to an increase in the purchasing power of current incomes. This indirect effect, operating through lower inflation, can also have a mitigating effect on poverty levels.

Liberalisation and Productivity

Productivity gains are the most immediate targets of the policy reforms that have centred around liberalisation and deregulation. Liberalisation can contribute to productivity gains in at least four different ways:

1) Liberalisation could result in reduction of *X-inefficiency*. This can happen in two ways: (a) a 'challenge-response' mechanism induced by a more competitive environment helps reduce the managerial slack; and (b) the existence of excess capacity can deter entry.

2) Liberalisation makes better exploitation of scale economies. These economies are possible through larger export markets.

3) Relaxation of import restrictions, or an increased ability to import due to increased exports, could then lead to better productivity performance due to the import of superior embodied technology and/or lower cost inputs.

4) Liberalisation makes possible faster rates of technological 'Catch up' in expanding sectors of comparative advantages.

Total factor productivity in Indian manufacturing was 0.61 per cent per annum during the period 1981-82 to 1990-91, dipped to 0.25 per cent per annum during 1991-92 to 1997-98, went down further to -0.09 per cent per year during 1998-99 to 2001-02 and then expanded sharply to 1.41 per cent per year between 2002-03 and 2007-08. The growth of productivity in manufacturing, therefore, follows a J-curve pattern, slowing down in the initial years after liberalisation, but picking up steam later.

Why is the curve J-shaped? Total factor productivity growth was slow during the early and mid-1990s because of the dramatic import liberalisation, the balance of payments deterioration and the exchange rate reform, all of which came as a shock to industry. Moreover, with the removal of import restrictions on consumer goods, certain types of capital goods were rendered obsolescent, which led to lower capacity utilisation in the late 1990s and early 2000s, and even lower productivity. As firms learned to cope and the benefit of new technologies and products became widespread, productivity improved dramatically from the mid-2000s.

Although other general measures on labour, land and infrastructure could improve total factor productivity somewhat, sector-specific factors are required to be analysed for taking necessary policy measures to take productivity growth in manufacturing to a higher level.

14.6 APPROACH TO A NEW INDUSTRIAL POLICY

The Government had released in recent past a 'Discussion Paper' outlining the basic contours of industrial policy suitable for rapid industrial growth. It presents vision, strategic objectives, and intent of the policy. These objectives and instruments to achieve the same are summarised below.

Clear Vision, Strategic Objectives and Intent

The policy aims to set a clear vision for the role of industry and industrial growth in the growth and development of the economy. A shared vision to develop a globally competitive Indian industry with skill and scale, which leverages technology will be developed through engagement with stakeholders. Strategic objectives have to be delineated with measurable outcomes. The policy has to also ensure that it embeds into itself sectoral objectives and provides an overarching umbrella policy framework. The timeframe for implementation of the policy needs to be decided taking into consideration the changing economic and business cycles of the world and the Indian economy, geopolitical trends and broad policy directions of the country. To begin with, the following strategic objectives are set forth for the policy, to enable commencement of work.

A) **Establishing Global Linkages**

1) India needs to strengthen global strategic linkages by– creating global brands out of India, strengthening linkages between Indian and global inductivity and intensifying FDI.

2) Brand building must gain importance alongside achieving quality and scale. The quantum of value addition must be increased at all levels. Larger the value addition, greater the positive externalities from economic activity.

3) FDI policy is largely aimed at attracting investment. Benefits of retaining investments and accessing technology in industrial sector have not been harnessed to the extent possible. FDI policy requires a review to ensure that it facilitates greater technology transfer, leverages strategic linkages and innovation

B) **Enhancing Industrial Competitiveness**

1) Competitiveness can be improved by reducing the cost of infrastructure such as power logistics causing regulatory compliance burden reducing the cost of capital and improving labour productivity.

2) Industrial infrastructure in India suffers from lack of funds and inefficiencies. Infrastructure financing relies heavily on banks due to the lack of developed debt and bond markets. Increased pressure on infrastructure from rapid urbanisation exacerbates inefficiencies leading to high cost of logistics.

3) Advances in technology are leading to emergence of new activities, major changes in existing systems and obsolescence at a rate faster than ever before. The central role of technology in next generation business has to be acknowledged and appropriated to ensure greater productivity and competitiveness.

4) In an open world, domestic tax structure and duty rates are an important factor in deciding the direction of flow of raw materials and final products. High direct tax rates and a duty structure that favours import of final products can act as disincentives for domestic manufacturers.

C) **Employing Gainfully a Growing Workforce**

India is now in the mid-point of the demographic dividend phenomenon which is expected to continue for another 15- 20 years. Demographic dividend has been a part of mainstream planning and policy making process in the last decades. There are several related concerns to name a few include the following:

- Some states would move out of the phenomenon earlier than others adding yet another dimension to existing inequalities
- Projected upward trends in automation and Artificial Intelligence (AI) leading to job losses
- Disproportionately slower growth in creation of jobs as compared to growth in output – the phenomenon of 'job-less' or 'ruth- less growth'.
- Poor outcomes in education, skill and health leading to an inability to harness the demographic dividend.

D) **Ensuring Sustainability and Responsible Industrialisation**

Industry is a major contributor to carbon emission and also has substantial resource footprint. It is also irrefutable that energy is a significant contributor to industrial growth as well as industrial emissions. Policies on utilisation of natural resources, including energy sources have to be aligned to support industrial growth. Sustainability has to be treated as integral to growth in all sectors and industry as a whole. A balance between industrial growth and improvement in environment and sustainability has to be achieved. Sectors that contribute to the latter should be given equal or more attention given that they also contribute to the former

E) **Enabling Ecosystem for Technology Adoption and Innovation**

1) India has the potential to diversify its strengths in the field of information technology. It is not just the largest software service provider, but can provide products and solutions and also become the 'Digital factory of the world', by becoming the vanguard of digital revolution. Despite foreign investments being received in the country over the last three decades, transfer of technology has largely remained at assembly level. Component manufacturing design and R&D activities have to be strengthened.

2) Right models of technology transfer need to be adopted to ensure that the transferred technology is enhanced and customised for Indian conditions. The issue of academia– research institutions industry linkages needs to be addressed. Across the board innovation should be promoted helping Indian firms increase their R&D spends and file high-quality patents that can be commercialised. Start-up ecosystem that plays a key role in this effort needs to be encouraged.

To sum up, there is a need to formulate an outcome-oriented actionable industrial policy that provides direction and charts a course of action for a globally competitive Indian industry which leverages skill scale and technology. Consultations will be held with industry bodies, industry captains, central government departments, state governments, think tanks, academia, and R&D institutes to understand perspectives of all stakeholders.

Check Your Progress 3

1) What was the critical issues facing the Indian industrial sector?

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2) What are the strategic objectives set forth for the new Indian industrial policy?

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14.7 LET US SUM UP

Industrialisation is the process by which an economy moves from primarily agrarian production to mass-produced and technologically advanced goods and services. This phase brings with itself an era of higher productivity, shifts from rural to urban labour, and increased standards of living. In India, post-independence, the national consensus was in favour of the rapid industrialisation of the economy as the key to economic growth and development. The Industrial Policy Resolution of 1948 marked the beginning of the evolution of the Indian Industrial Policy. The Resolution emphasised the importance to the economy of securing a continuous increase in production and its equitable distribution, and pointed out that the State must play a progressively active role in the development of Industries. In the subsequent years, India's Industrial Policy developed through successive Industrial Policy Resolutions and Industrial Policy Statements. The key role in industrial development programme was in the public sector. The aim was to make the economy self-sustaining in producers' goods industries such as steel, machine building, etc., so that the quantum of external assistance needed could be curtailed to a very low level. Later in the mid-eighties, the trend towards liberalisation became more pronounced. Industrial policies took a shift from predominantly Socialistic pattern in 1956 to Capitalistic since 1991. Industrial policy 1991 onwards dealt with liberalising licensing and measures to encourage foreign investments. However, the major challenges that have restricted industrial growth inter-alia include– inadequate infrastructure, restrictive labour laws, complicated business environment, slow technology adoption, low productivity,